

ML Assignment 1 - Polynomial Regression

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1 Introduction

This report documents the development of polynomial regression models for two distinct geothermal energy engineering problems. As specified in the assignment requirements, the solutions strictly utilize polynomial regression to predict the continuous target variables. This was implemented using scikit-learn's `PolynomialFeatures` to generate the polynomial terms, combined with `Ridge` regression for regularization. Model selection and hyperparameters were evaluated using 5-fold cross-validation.

2 Phase 1: Power Plant Steam Turbine Optimization (var1)

2.1 Problem Overview

The objective of Phase 1 is to optimize a multi-stage steam turbine system for a surface energy generation facility. The turbine efficiency is dependent on six operational parameters (x_1 through x_6), representing percentage deviations in valve adjustments, flow rates, and pressures. The target variable is the Net Power Score (y).

2.2 Model Selection

Polynomial features were generated using all 6 operational parameters. Degrees ranging from 2 to 6 were rigorously evaluated. To mitigate multicollinearity and control overfitting in high-dimensional polynomial spaces, Ridge regression was applied.

A polynomial of degree 4 with Ridge regularization ($\alpha = 2.0$) was selected as the final model. It provided the optimal balance between bias and variance, achieving a high Cross-Validation (CV) R^2 while minimizing the CV Mean Squared Error (MSE). The selected polynomial of degree 4 evaluates 210 terms.

Features	Degree	Terms	Train R^2	Train MSE	CV R^2	CV MSE
All 6	2	28	0.7413	2.8339	0.7214	3.0198
All 6	3	84	0.9258	0.8126	0.9074	0.9902
All 6	4 (Final)	210	0.9653	0.3807	0.9393	0.6521
All 6	5	462	0.9875	0.1375	0.9589	0.4439

Table 1: 5-fold cross-validation results for Phase 1 (var1) using Ridge ($\alpha = 2.0$).

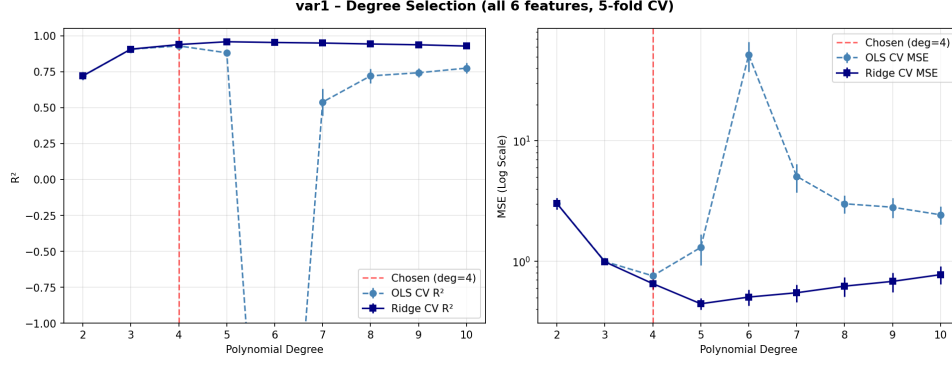


Figure 1: Phase 1: CV R^2 and MSE vs. Polynomial Degree.

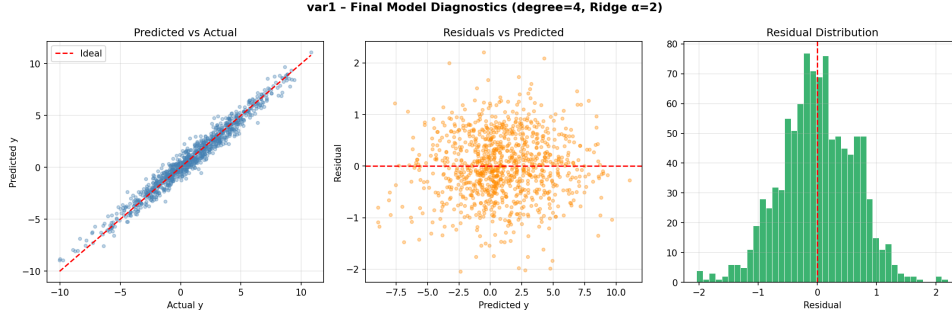


Figure 2: Phase 1: Final Model Diagnostics (Residuals and Predictions).

3 Phase 2: Subterranean Thermal Reservoir Mapping (var2)

3.1 Problem Overview

Phase 2 requires mapping a geothermal survey block to select exact coordinates for drilling new extraction wells. The dataset comprises 3D spatial coordinate offsets (x_1 : East-West, x_2 : North-South, x_3 : Vertical depth). The target variable is the Thermal Anomaly Score (y), which reflects complex geological structures.

3.2 Model Selection

Models incorporating all three spatial coordinates were extensively tested across polynomial degrees 2 through 12. Cross-validation indicated that predictive performance steadily improved up to degree 10. Beyond degree 10, the model began exhibiting signs of overfitting (CV MSE plateaued and R^2 slightly degraded). Therefore, a degree 10 polynomial (yielding 286 terms) with Ridge regularization ($\alpha = 0.05$) was selected.

Features	Degree	Terms	Train R^2	Train MSE	CV R^2	CV MSE
x_1, x_2, x_3	4	35	0.9296	3.1792	0.9223	3.5104
x_1, x_2, x_3	6	84	0.9906	0.4266	0.9875	0.5625
x_1, x_2, x_3	8	165	0.9963	0.1665	0.9938	0.2761
x_1, x_2, x_3	10 (Final)	286	0.9969	0.1393	0.9945	0.2483
x_1, x_2, x_3	12	455	0.9978	0.1002	0.9942	0.2578

Table 2: 5-fold cross-validation results for Phase 2 (var2) using Ridge ($\alpha = 0.05$).

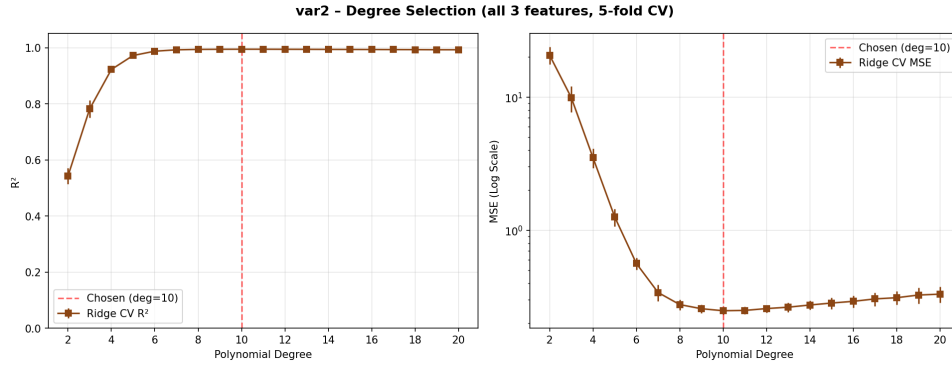


Figure 3: Phase 2: CV R^2 and MSE vs. Polynomial Degree.

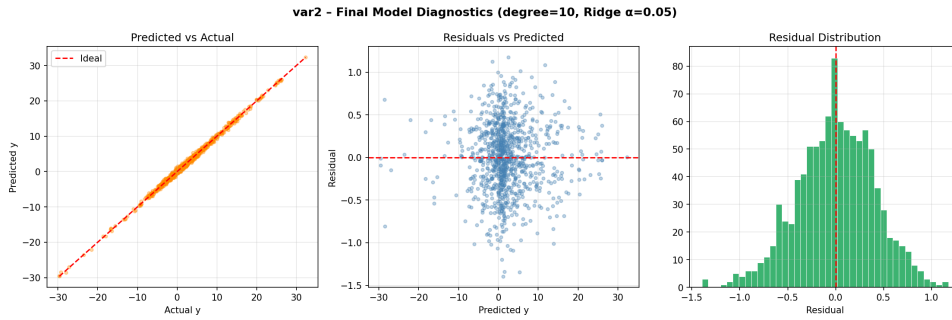


Figure 4: Phase 2: Final Model Diagnostics (Residuals and Predictions).

4 Deliverables

- BT2024268_pred_var1.csv: Test CSV file containing predicted values for y (Net Power Score).
- BT2024268_pred_var2.csv: Test CSV file containing predicted values for y (Thermal Anomaly Score).
- BT2024268_polynomial_regression.py: Self-contained script for training models and generating predictions.
- **GitHub Link:** [Link](#)